

Method

We replicate the joystick-policy finetune schedule from Appendix §C of *Demonstrating MuJoCo Playground* [1], applied to the Robotis OP3 platform. Three PPO seeds (1, 7, 42) of the upstream Op3Joystick env were trained for 100 M steps at the published default config (*stage 1*; paper §C.25.a step 1); each was then finetuned for 50 M with the yaw command range expanded from ± 0.7 to $\pm 2\pi$ rad/s (*stage 2a*; paper §C.25.a step 2 cut at half budget) and continued by another 50 M of the same finetune (*stage 2b*; cumulative training 200 M per seed). Step 3 of the paper schedule (rough-terrain finetune) is not applicable: the Playground codebase exposes a single Op3Joystick env at the pinned commit (paper Table V), with no rough-terrain variant — in contrast to Unitree Go1, G1, T1 and Berkeley Humanoid, each of which has flat- and rough-terrain training environments. Hyperparameters otherwise verbatim Playground locomotion defaults.

We then replayed three 25-second deterministic command scripts on each of the nine resulting policies in MuJoCo: a *yaw sweep* (slow ± 0.5 / fast ± 1.5 rad/s), a *vx sweep* (slow ± 0.3 / fast ± 1.0 / -0.5 m/s, asymmetric to match the asymmetric training distribution), and a *vy sweep* (slow ± 0.3 / fast ± 0.6 m/s). Each script visits four active commands separated by 1.5-second stops to expose transition behaviour. Per-segment statistics: mean tracking error normalised by commanded magnitude — $\|\text{cmd_xy} - \text{vel_xy}\| / |\text{cmd}|$ for linear axes, $|\text{cmd_yaw} - \omega_z| / |\text{cmd_yaw}|$ for yaw — and falls detected by the env’s training-time termination thresholds (gravity $z < 0.85$ or torso $z < 0.21$ m).

Sample size. This is an exploratory run with three seeds per training stage. The numbers below describe what we *observed in this particular run*, not estimates of population behaviour; with $n = 3$ we can only loosely characterise tendencies and report the seed-to-seed variation we saw. A larger seed sweep would be needed to make any claim about the curriculum’s typical effect.

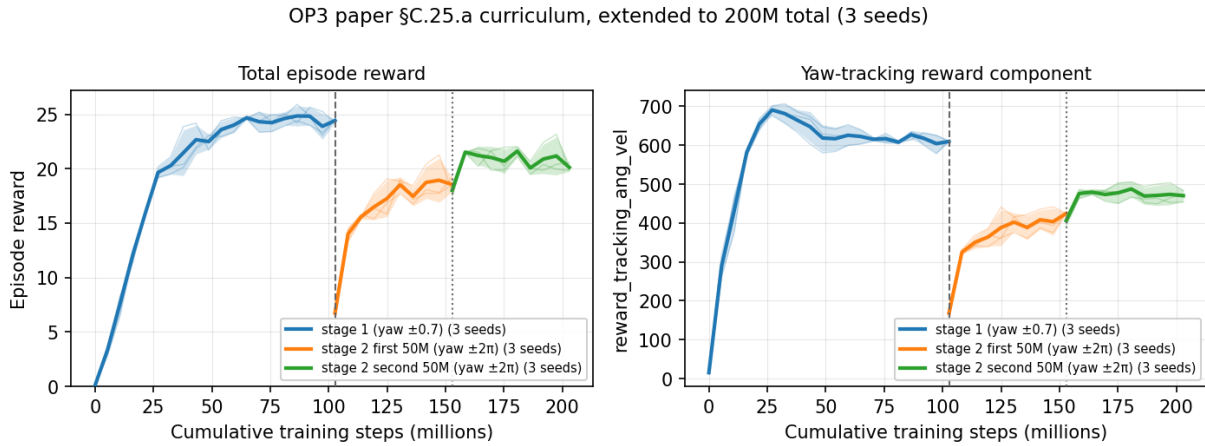


Figure 1: Curriculum reward curves over 200 M cumulative training, 3 seeds (mean $\pm \sigma$ band, thin per-seed lines underneath). Vertical dashed line: stage 1 $\rightarrow$ stage 2a (yaw range $\pm 0.7 \rightarrow \pm 2\pi$). Dotted: stage 2a $\rightarrow$ stage 2b. Left: total reward; right: `reward_tracking_ang_vel` component.

Results

Yaw — in this run, the curriculum looks like a trade-off rather than a strict improvement. Tracking error per yaw segment (% of commanded):

Segment	Cmd									
	(rad/s)	s1·st1	s1·st2a	s1·st2b	s7·st1	s7·st2a	s7·st2b	s42·st1	s42·st2a	s42·st2b
Slow +yaw	+0.5	22%	46%	50%	36%	71%	75%	20%	35%	26%
Slow –yaw	–0.5	18%	39%	26%	22%	95%	62%	16%	48%	44%
Fast +yaw	+1.5	11%	7%	8%	8%	9%	12%	13%	17%	11%
Fast –yaw	–1.5	15%	8%	6%	14%	12%	11%	13%	11%	8%

In our three seeds, stage 2 reduced fast-yaw error by 1–7 pp and increased slow-yaw error by 4–73 pp. The trade direction was the same across the three runs, with magnitudes varying considerably. The boldfaced 95% (s7 stage 2a, slow –yaw) corresponds to a policy whose body does not rotate in response to a 0.5 rad/s command — a lean-left/lean-right oscillation visible in the recording. The second 50 M (stage 2b) partially recovered slow-yaw quality on each of the three seeds (most strikingly on s7, 95% → 62%) without further changing the fast-yaw side; no seed returned to stage-1 levels on slow yaw.

Forward / backward (vx) — pronounced asymmetry in our sample, fall-prone at fast forward. Each cell shows tracking error and falls in segment.

Segment	Cmd (m/s)	s1·st1	s1·st2	s7·st1	s7·st2	s42·st1	s42·st2
Slow forward	+0.3	99%, 0	98%, 0	99%, 0	99%, 0	98%, 0	59%, 0
Slow back	–0.3	99%, 0	99%, 0	99%, 0	99%, 0	99%, 0	99%, 0
Fast forward	+1.0	74%, 1	71%, 0	51%, 0	46%, 1	85%, 2	59%, 1
Fast back	–0.5	26%, 0	28%, 0	19%, 0	24%, 0	21%, 0	25%, 0

Three patterns recurred across the six policies we tested. (i) At slow vx (± 0.3 m/s) the policies appeared to under-respond — five of six slow-forward segments and all six slow-backward segments showed >98% tracking error, despite ± 0.3 being within the training range $[-0.6, 1.5]$. (ii) Backward at -0.5 m/s was the cleanest segment of the matrix (19–28% error, no falls observed) even though the magnitude sits near the trained negative range edge. (iii) Forward at $+1.0$ m/s tracked substantially worse (46–85% error) and produced 1–2 falls per 4 s segment in 4 of 6 policies. The asymmetric training range did not yield symmetric robustness in our sample.

Strafe (vy) — the most stable axis in our sample.

Segment	Cmd (m/s)	s1·st1	s1·st2	s7·st1	s7·st2	s42·st1	s42·st2
Slow + (left)	+0.3	24%, 0	26%, 0	23%, 0	27%, 0	23%, 0	39%, 0
Slow – (right)	–0.3	27%, 0	27%, 0	35%, 0	37%, 0	23%, 0	24%, 0
Fast + (left)	+0.6	30%, 0	31%, 0	29%, 0	30%, 0	28%, 0	38%, 2
Fast – (right)	–0.6	25%, 0	26%, 0	28%, 0	30%, 0	26%, 0	26%, 0

Strafe was the most consistent axis we sampled: no falls in any stage-1 strafe segment, and 2 falls (one seed at fast +vy) across all six stage-2 strafe segments. The yaw curriculum cost 1–10 pp of strafe precision in our sample without buying observable benefit on this axis.

Seed 7 behaved as an outlier in our sample. Of the three stage-2a runs, seed 7 ranked first by every training-eval indicator we logged: highest total reward (19.21 vs 18.52 / 17.89), highest reward_tracking_ang_vel (454 vs 425 / 423), longest mean episode length (854 vs 839 / 823), and smallest termination penalty (−0.27 vs −0.34 / −0.30). Behavioural inspection ranked it last: under a 0.5 rad/s yaw command its body leaned left and right rather than turning. The disagreement persisted across the second 50 M of training — seed 7’s reward_tracking_ang_vel was unchanged from stage 2a to stage 2b (454 → 454) while its slow-yaw error fell 33 pp (95% → 62%). One possibility is that the yaw-tracking reward ($\exp(-(\text{cmd}_\omega - \omega_z)^2 / \sigma)$, Table VI of [1], a per-step rate gap) does not always penalise lean-style behaviour because it scores how well the IMU rate signal averages to the commanded value rather than how much the body accumulates angular displacement; another is that this is simply one anomalous run. With three seeds we cannot tell which. A wider seed sweep is needed before drawing any conclusion about training-eval reward as a deployment-selection criterion in this setup.

References

- [1] Kevin Zakka et al. “Demonstrating MuJoCo Playground”. In: *Proceedings of Robotics: Science and Systems*. Los Angeles, CA, USA, June 2025. DOI: [10.15607/RSS.2025.XXI.020](https://doi.org/10.15607/RSS.2025.XXI.020). URL: <https://www.roboticsproceedings.org/rss21/p020.html>.